

Assignment 5 - Timers

25.2.2024

Sent 100%

Forsøk 1



Se gjennom tilbakemelding

28.2.2024

Forsøk 1 Poengsum:

100%



Legg til kommentar

Anonym vurdering: **nei**

Ubegrenset antall forsøk tillatt

16.12.2023 til 29.4.2024

▼ Detaljer

Preparations

This assignment assumes you have completed all steps of the previous assignments. The resources and general instructions from previous assignments are still useful here, this assignment will only list new relevant resources.

1. Create a new program for target DISCO-L475VG-IOT01A (B-L475E-IOT01A) under your git folder (*<course folder>/assignments/*) called *assignment_5*.

Resources

- [Mbed OS documentation](https://os.mbed.com/docs/mbed-os/v6.16/apis/index.html)  (<https://os.mbed.com/docs/mbed-os/v6.16/apis/index.html>)
- Lecture notes

Requirements

In this assignment you will create an egg timer. Buttons are used for input, the [piezo buzzer](https://www.adafruit.com/product/160)  (<https://www.adafruit.com/product/160>) (from the parts kit) is used to sound the alarm and the LCD will show the remaining time.

The default timer is 1 minute. The timer should start paused.

1. 10% Connect four buttons to digital inputs. You can use these with [interrupts](https://os.mbed.com/docs/mbed-os/v6.16/apis/interruptin.html)  (<https://os.mbed.com/docs/mbed-os/v6.16/apis/interruptin.html>) or with regular DigitalIn, that's up to you
2. 10% Connect the piezo buzzer to a [PWM output](https://os.mbed.com/docs/mbed-os/v6.16/apis/pwmout.html)  (<https://os.mbed.com/docs/mbed-os/v6.16/apis/pwmout.html>) (this way you can control the tone it plays)
3. 10% Connect the RGB LCD to I2C1 and VCC/GND. (see underside of the LCD)
4. 10% Show the remaining time on the LCD
5. 10% Button 1 should start/pause the timer

6. 10% Button 2 should reset the timer back to 1 minute and pause it
7. 10% Button 3 should add 5 seconds to the timer
8. 10% Button 4 should subtract 5 seconds from the timer
9. 20% Stop the countdown when the timer reaches 0 and sound the buzzer.

The countdown is best executed using the *Timeout* class from Mbed OS, avoid the *Timer* class.

Delivery

Deliverer the following as text in Canvas:

- Link to Bitbucket where to code has been delivered.
Directly to the file if there is only 1, or the directory if more than 1.
- Link to YouTube (or other video service) with video showing each of the requirements.
Make sure that each requirement you have completed is clearly shown.
You don't need to show the actual code, just which of the requirements are met.